

# Retail Business Performance & Profitability Analysis

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## Abstract

This project analyzes transactional retail data from the Sample Superstore dataset to identify profit-draining product categories, evaluate the impact of order fulfillment speed on profitability, and uncover the true driver of margin erosion. Using SQL, Python, and Power BI, the analysis finds that discount levels — not shipping speed or region — are the primary cause of profit loss, with discounts above 20% consistently producing negative profit across the dataset.

## Introduction

Retail businesses often struggle to identify which products or pricing strategies quietly erode profit even while sales volume looks healthy. This project was undertaken to analyze the Superstore transactional dataset (9,994 orders, 2014–2017) and answer three questions: which product categories drain profit, whether slower order fulfillment correlates with lower profitability, and how discounting practices affect the bottom line. The goal was to derive clear, actionable insights that a retail business could use to adjust pricing and inventory strategy.

## Tools Used

- **SQL (SQLite via DB Browser):** Data cleaning, profit margin calculations by category/sub-category, regional and discount-band aggregation
- **Python (Pandas, Seaborn, Matplotlib via Google Colab):** Date parsing, fulfillment-time (days-to-ship) correlation analysis, visualization
- **Power BI Desktop:** Interactive dashboard with KPI cards, profitability charts, regional treemap, and quarterly trend analysis
- **Tableau Public:** Cross-validation dashboard to confirm findings across an independent BI tool

## Steps Involved in Building the Project

- **Data Cleaning (SQL):** Imported the Superstore CSV into SQLite, checked for null values and duplicate line items, and created a cleaned table for analysis.
- **Profitability Analysis (SQL):** Calculated profit margins by category, sub-category, and region; segmented orders into discount bands (0%, 1–20%, 21–40%, 40%+) to test discount impact.
- **Fulfillment Correlation (Python):** Parsed mixed-format order/ship dates with Pandas, computed days-to-ship per order, and ran Pearson correlation against profit, both overall and by category.
- **Dashboard Build (Power BI & Tableau):** Built KPI cards (Total Sales, Profit, Orders, Avg Discount, Profit Margin), a sub-category profit chart, a regional treemap, a quarterly sales/profit trend line, and a discount-band profit chart.

## Conclusion

The analysis identified **Tables and Bookcases** as consistently loss-making sub-categories despite healthy sales volume. Fulfillment speed showed **negligible correlation with profit** (near zero across all categories), ruling out shipping delays as a cause of lost margin. The clearest finding was the **discount-band analysis**: orders with 0–20% discount remained strongly profitable, while orders with 21%+ discount turned profit negative, with the 40%+ band showing the steepest losses. The data also revealed a seasonal pattern — Q4 sales nearly quadrupled over Q1, but profit only doubled, indicating heavier holiday-season discounting eats into margin growth.

**Recommendation:** cap discounts at 20% for Furniture sub-categories (particularly Tables and Bookcases) to restore profitability without sacrificing sales volume.